Curriculum Vitae

# SeokHyeon (Lucas) Kong

Seoul, South Korea | lucasshkong@gmail.com | lucasshkong.github.io

## RESEARCH INTERESTS

Quantum Computer Architecture, Quantum Compiler, Quantum Error Correction and Detection, Quantum Controller, Quantum Programming Language, Quantum Algorithm Optimization

## EDUCATION

**Sungkyunkwan University (SKKU)** — **Suwon, South Korea**
*College of Information and Communication Engineering* — *Expected February 2026*
Bachelor of Science in Electronic and Electrical Engineering — **GPA: 3.95/4.0 (Major: 4.0/4.0)**
Bachelor of Science in Advanced Semiconductor Engineering — System Architecture Track

**The Pennsylvania State University** — **University Park, PA**
*School of Electrical Engineering and Computer Science (EECS)* — *Fall 2024*
Exchange Student in Computer Engineering — **Grade: 4.0/4.0**

## PUBLICATIONS & PRESENTATION

**S. H. Kong**, D. Kim, K. Maeng*, and E. Seo*, "Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy," *IEEE Computer Architecture Letters*, under review, 2025

**S. H. Kong**, C. Park, J. Moon, and D. Dahl*, "Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices," *Poster presented at the Quantum Information Research Support Center (Qcenter)*, Suwon, South Korea, Sep. 2025

**S. H. Kong** and H. Oh*, "Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system," *Journal of the Korean Institute of Information and Communication Engineering*, vol. 27, no. 10, pp. 1208–1213, Oct. 2023

## RESEARCH EXPERIENCE

**Research Mentee** — **Remote**
*IonQ, Advised by Dr. Denny Dahl* — *08/2025–09/2025*

- Addressed the Instant Insanity problem on NISQ devices by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods.
- Applied a Variational Quantum Algorithm to the Four Corners Map Coloring problem.

**Undergraduate Research Intern** — **Suwon, South Korea**
*ArchLab @ SKKU, Advised by Prof. Dongmoon Min* — *07/2025–08/2025*

- Built foundational expertise in Quantum / Cryogenic Computer Systems through literature reviews, technical presentations and academic defenses.
- Explored cryogenic computer architecture - processors, interconnects, caches, and memory - as well as quantum computer systems with a focus on superconducting qubits.

**Undergraduate Researcher** **University Park, PA**
*Penn State CSE, Advised by Prof. Kiwan Maeng* *08/2024–09/2025*

- Conducted in-depth GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differentially private training.
- Proposed an optimization that resolved the dominant Geometric operation overhead, reducing it from 50% to 19% while preserving Gaussian noise quality.

**Undergraduate Research Intern** **Pohang, South Korea**
*POSTECH QCQN (Quantum Computing and Quantum Networks)* *01/2025*

- Performed 397 nm fluorescence (Doppler) and 729 nm quadrupole spectroscopy on $Ca^+$ ions and enhanced system stability through targeted adjustments.
- Implemented micromotion optimization via 2D scanning with DC compensation rods.

## HONORS & AWARDS

Quantum Hackathon - Director's Award *2025*

President's List (Honor Society at SKKU, **2 consecutive years**) *2024, 2025*

Dean's List (**5 consecutive semesters**) *SP2023, FA2023, SP2024, FA2024, SP2025*

Korea-U.S. Student Exchange Scholarship in the field of high-tech industry ($9,000) *FA2024*

## TEACHING EXPERIENCE

**Academic Tutor** **Suwon, South Korea**
*Sungkyunkwan University (SKKU), Intro to Electromagnetism* *Fall 2023*

**International Summer School Teaching Assistant** **Seoul, South Korea**
*Sungkyunkwan University (SKKU), Engineering Mathematics 2* *Summer 2023*

## TECHNICAL SKILLS

**Software:** Qiskit, Python, C, CUDA, MATLAB, LaTeX
**Hardware:** System Verilog, Verilog, Verdi, Virtuoso, Nsight Systems, Nsight Compute

## CERTIFICATION

Qiskit Global Summer School 2025 – Quantum Excellence, IBM Quantum

## REFERENCES

**Kiwan Maeng, Ph.D.**
*Assistant Professor, Computer Science and Engineering, The Pennsylvania State University*
*Email*: kvm6242@psu.edu

**Denny Dahl, Ph.D.**
*Customer Success Director, IonQ*
*Email*: denny.dahl@ionq.co